PROPOSAL

TECHNICAL SUPPORT FOR THE DEVELOPMENT OF A NET-ZERO TRANSFORMATION PATHWAY THROUGH A RESOURCE AND ENERGY EFFICIENCY CHECK FOR THE COISHCO PLANT, AUSTRAL GROUP S.A.A PERÚ

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Table of content

1 The Contractor

The contractor is the Institute for Applied Material Flow Management (*Institut für angewandtes Stoffstrommanagement* – hereinafter IfaS) who shall execute the project on behalf of the Trier University of Applied Sciences. IfaS is a higher research institution of the Trier University of Applied Sciences located at the Environmental-Campus Birkenfeld.

1.1 Institute for Applied Material Flow Management (IfaS)

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IfaS was founded in 2001 at the Trier University of Applied Sciences located at the Environmental Campus Birkenfeld (Germany). As a university-based research institute, IfaS' main objective is to integrate and strengthen sustainable structures at industrial, communal and regional levels, in national, EU-wide, and international contexts. IfaS is one of the very few institutions that successfully apply regional material and energy flow management (MFM) assisting industry, communities, regions and countries in developing ***Zero Emission*** and ***Circular Economy*** concepts. The unique working field of IfaS is to analyse, mobilise, optimise and develop new and innovative material, energy, and financial flows in industries and regions, in order to create added value for all regional stakeholders, enhance the business competitiveness and legal compliance of industrial partners, and also reduce negative environmental impacts.

Over one-and-a-half decades, IfaS has carried out its work successfully through the implementation of proven and innovative environmental technologies and services on a sound economic basis as well as the setting up, managing, and maintaining networks of key stakeholders, especially in energy and resource management related issues. The interdisciplinary team of IfaS, including the Managing Director, Professor Dr. Peter Heck, and eight professors as the board of directors, consists of about 75 permanent staff members and a further 40 visiting researchers from various scientific disciplines to develop and execute numerous regional, national, and international projects points in Asia, Asia Pacific region, Central and South America, Africa and Europe. The project portfolio ranges from single energy efficiency improvement projects for industry, towards integration of renewable heating and electricity systems based on regionally available renewable energy sources to holistic national zero emission concepts and strategies. IfaS can draw on a far-reaching and comprehensive network of industrial partners (innovative environmental technology providers from Germany), international research organisations and non-governmental institutions as well as political institutions and organisations on German and international levels.

1.2 IfaS International Network

The development of new business models integrating climate protection, resource and energy efficiency, and the strengthening of local economies on a global level, is the challenge handled by the International Project Management team of IfaS. Therefore, its task is to introduce knowledge and technology transfer, adapt technical concepts to local needs, develop innovative financing concepts, and support the implementation of projects. Based on the principle premise "Discover Potentials - Optimise Processes - Create Regional Added Value", IfaS is conducting within the framework of material flow management, Circular Economy and zero-emission-concepts detailed material and energy flow analyses, which form the basis for holistic system optimisation, the presentation of energy and resource efficiency potentials, renewable energy generation and environmental and energy management systems as well as the calculation of Greenhouse Gas (GHG) emission reduction potentials. Targeted trainings abroad and in Germany for international decision makers facilitate the initiation and realisation of innovation in developing and emerging countries.

IfaS has the ability to leverage additional value such as ancillary insight, knowledge, intellectual property, networks, national political influence, further academic and practical education in all of its projects. As IfaS is concerned with the development of Zero Emission, and Circular Economy concepts all across national and international levels, IfaS holds a substantial network of IfaS integrated cooperation partners all over the world.

Through the close collaboration with highly specialised and innovative environmental technology providers and planning experts, IfaS not only offers consultancy services to execute the RECs (and following in-depth assessments) with its in-house personnel resources but also provides its assessments based on best-available technologies on actual market prices in close cooperation with the technology service providers. On-demand, IfaS could rely on its company network with more than hundred specialised environmental technology providers to provide additional technical advice and consultancy services – even on short-term notice – to a range of topics which were initially not tasked at the initial conference call but emerging during the conduction of the RECs advisory services.

Thanks to the international network of alumni from the institute's own study course IMAT (Master in International Material Flow Management)[1], IfaS (in addition to the German environmental service providers) has access to various companies in numerous countries, whereby the MFM approach and the entire circular economy methodology is already known and applied.

[1] For more information on the master program, please refer to http://www.imat-master.com/.

1.3 Established track record of similar consultancy services

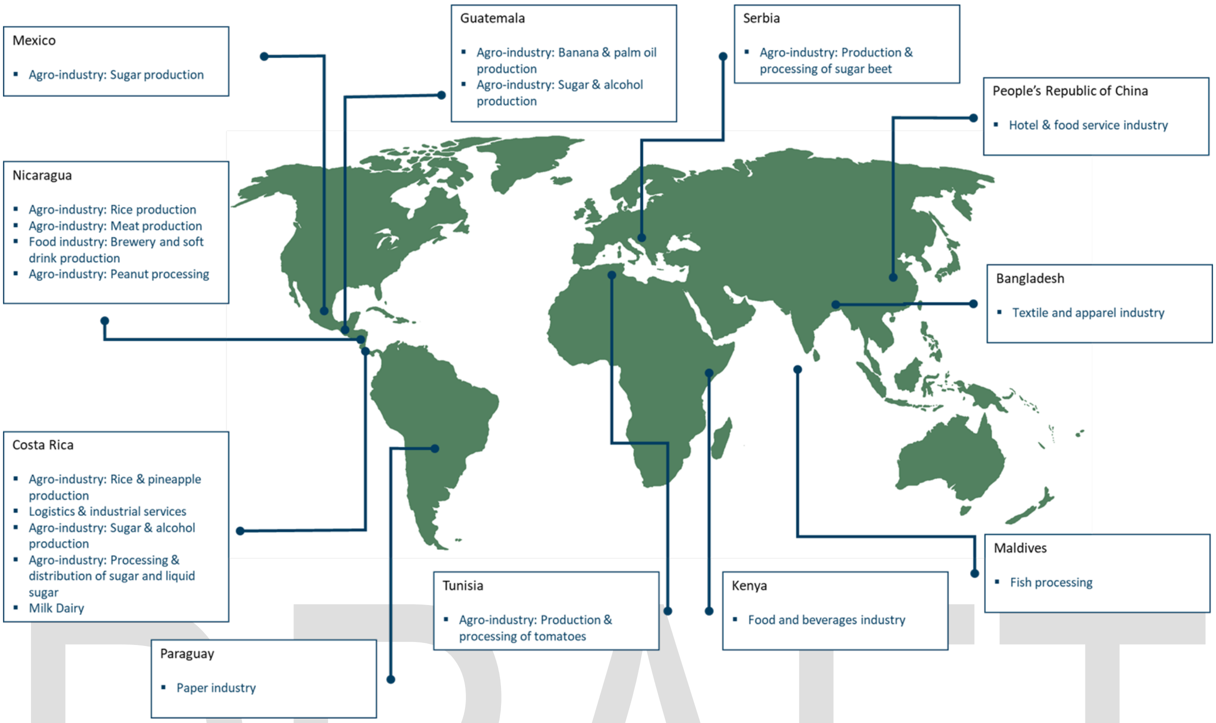
IfaS, together with its broad network of specialists from reliable partner companies worldwide, has a long and solid track record in the designated advisory service areas. The technical assistance spans into four work packages (WP) and encompassed the following areas:

- WP1) Energy efficiency assessment of the production process and installed cross-sectional technologies
- WP2) Identification, optimisation and utilization of biomass material flows
- WP3) Improvement of efficiency of water use in the production plant
- WP4) Identification of appropriate wastewater treatment technologies

The REC advisory services follow typically a five-step methodological approach for each potential project identified, namely:

1) Identification and definition of relevant areas to be investigated (boundary definition), in close consultation with the client,
2) Data procurement towards Material Flow Analysis [MFA] to determine the status quo (based on the qualitative and quantitative evaluation of material and energy flows),
3) Identification of optimisation potentials in order to apply efficient and effective technologies and management approaches, including a description of the selection of best available technologies [BAT] and processes,
4) Economic pre-feasibility analysis of the identified improvement potentials/projects,
5) Evaluation of environmental impacts (such as GHG abatement potential).

Figure 1: Conducted REC advisory services by IfaS



2 Work Plan of the Contractor

2.1 Objectives of the technical support services

The objective of the consultancy services is to provide technical support for the development of a net-zero transformation pathway for the production plant Coishco of the Austral Group S.A.A. in Perú through a Resource and Energy Efficiency Check (REC). The decarbonisation plan will consider the following aspects:

- Status quo analysis of Coishco production plant, its existing equipment and processes to determine the baseline consumption of resources and energy.
- Estimation of the baseline inventory of carbon emissions (Scope 1 and Scope 2 according to GHG Protocol).
- Execution of a technical site visit to the production plant to identify significant energy consumers, evaluate energy and resource management practices and to understand current production processes. The technical site visit will be prepared based on questionnaires, interviews and review of documentation and/or existing studies.
- Identification and definition of recommendations as optimization measures for loss reduction, mitigation of carbon emissions, energy and resource related cost savings, and implementation of renewable energies (i.e. solar energy for electric and thermal energy generation). The recommendations should include their technical and economic pre-feasibility analysis.
- Based on the identified optimization measures, development the net-zero transformation pathway according to the Client's strategy.

- Definition of a system of indicators in order to measure, control, manage and optimize the consumption of energy and water, as well as, generation of waste and wastewater. The system will be able to respond to operational eventualities and to optimize the management of energy and resources.
- Evaluation, categorization and prioritization of recommended measures in terms of costs, reduction of emissions, in the medium and long term.
- Definition of strategies, recommendations and corporative policies for the implementation of the net-zero transformation pathway.

2.2 Tasks of Advisory Services

The work plan featuring the methodology and scope of the contractor for the designated **REC** addresses the **four key strategic advisory service areas.** Based upon a review of existing documents, meetings with relevant staff and contractors, site visits and (if necessary) workshops, the Consultant will perform the following tasks for the **REC**:

- **Task 1)** Preparation of the REC advisory services and the onsite visit to the client's installations. This task consists mainly in the acquisition and review of relevant data and studies, which includes:
  i. Evaluation of energy efficiency measures previously identified by the client.
  ii. Review of the evaluation matrix for the prioritization of identified projects.
  iii. Understanding of the objectives, actions and goals set by the client's Energy Efficiency Committee.
  iv. Support in filling out the preparation questionnaire for the initial analysis of the energy consumption behaviour and the initial efficiency check of cross-sectional technologies.
- **Task 2)** Analysis of the status quo by using the IfaS´ material flow analysis [MFA] developed for Circular Economy project design to document current resource and energy consumption (and create system transparency). This task includes:
  i. Identification and definition of relevant areas to be investigated (boundary definition), in order to lay clearly down the focus of each consultancy for every individual REC.
  ii. Material Flow Analysis to determine the status quo (based on a qualitative and quantitative evaluation of material and energy flows), including:
     - *Material and energy-based Input-Output Analysis*
     - *Associated GHG emission starting inventory (related to manufacturing and logistic activities) based on the GHG Protocol for Corporates and UNFCCC approach;*
     - *Process flow diagrams ("Maps") for material (raw material, biomass, etc.), energy (electricity, fuel, heat, etc.) and water (and waste water);*
     - *Definition of Compound Annual Growth Rate (CAGR) and associated GHG trajectory within the Business as Usual (BAU) scenario;*

- *Determination of Levelized Cost of Services for each tier (e.g. levelized cost of water, carbon adjusted levelized cost through internalization of external costs (such as carbon) into single services);*
  - *Levelized cost is a value of the average net present cost of a service over its lifetime. The levelized cost approach become popular for the calculation of electricity (LCoE) on a full-cost basis and can be applied for each kind of service (such as heat, cold, water, waste water, etc.);*
  - *IfaS uses the concept to identify sensitivities (or hot spots) in order to derive parameters if or at which point decarbonization activities are economically relevant (possible, conducive, indispensable), for instance: Heavy Fuel Oil price increase about 5% p.a. for 3 consecutive years at contemporaneous reduction of CAPEX for renewable alternatives about 2.5% p.a.;*

- **Task 3)** Identification of opportunities (potential analysis) to optimise the efficient utilisation of energy and resources by clustering the potential towards theoretical, technical and economic potentials. The task will focus on measures to reduce the consumption of fossil fuels (e.g. Heavy fuel oil in stem boilers) This task includes:

  i. Description and selection of best available (process) technologies (BAT) and management approaches to capitalise the identified optimisation potentials;

  ii. Single business plans and economic evaluation (with key performance indicators) on single project level as well and on aggregated company level for all identified opportunities;

  iii. Critically scrutinize fossil-based industry-operated power plants and development of alternative concepts towards replacement on the basis of renewable technologies;

  iv. Support achievement of sustainable transformation and climate resilient operations in line with the SDGs within the company's internal operations, supply chains, and the environments.

- **Task 4)** Evaluation of carbon mitigation potential and development of net-zero transformation and water sensitive pathway. This task includes:

  i. Development of a decarbonisation pathway (net-zero carbon trajectory) based on the identified energy and resource saving options. The trajectory is developed till the companies set net-zero target year including expected CAGR effects.

  ii. Evaluation of carbon mitigation, carbon adaptation potential and determine Levelized Cost of Abatement per investment in EUR/t CO_2e

  iii. Evaluation of carbon sink options (e.g. protection of carbon rich biome through abandon and decommission, re- and afforestation, built-up of stable humus compounds and biochar application [particularly for agri-businesses with own cultivation) to achieve carbon neutrality within the company balancing/offsetting nonavoidable carbon emissions.

  iv. Water stewardship-based risk assessment both for water and waste water management within the company's operational control. Definition of associated GHG abatement potential in energy autarkic (or energy+) waste water and sewage sludge management.

v. Quick-check of NDC compliance for existing and envisaged resource and energy related activities.

- **Task 5)** Preparation of a standardised report, which includes the main assumptions, economic and technical calculations, recommended actions and respective cost-benefit estimations as well as prediction of GHG abatement potentials.

The general workflow of the REC will be explained below. The aforementioned tasks are explained in detail in the subsequent sub-chapters.

2.3 Proposed Workflow of REC

The project starts with the preparation phase. Firstly, IfaS will send its REC-proven questionnaire to create an initial data information base and an overview on the clients´ material and energy demand to identify the available data as well as their data gaps. Furthermore, in advance to the on-site visit and survey of the Client's installations, regular communication via Email, and Video-Conference shall be implemented between the client and the consultant to built-up a reciprocal understanding and to avoid misunderstanding. Once the IfaS´ questionnaire is completed and all relevant data and studies have been provided by the client, the available data are analysed by the consultant for their completeness and accuracy. Within ten days after the reception of relevant data, the first personal contact of the client and the consultant is established via conference call with the client's assigned contact person. Herein the provided data are revised and relevant areas to be investigated within the REC are determined (system boundary). Once the site-visit agenda and administrative issues (e.g. entry permit/visa) are cleared bilaterally between Consultant and Client, the onsite inspection *[serving as the* ***"Site Visit"****]* will be conducted.

The site visit starts with the **initial conception meeting (ICM)** on the first day attended by the leading representatives of the client. The following topics will be presented by the Consultant (or consultancy team) within the ICM to the client:

1. General overview (presentation) on the regional/industrial material flow management method, aims and objectives with regard to the project and the client's business
   a. Material and Energy Flow Analysis
   b. Results of the evaluated energy efficiency measures previously identified
   c. Initial GHG emissions baseline
   d. Basic aspects of cash flow analysis and business planning
   e. Project management plan for the execution of the REC onsite visit
2. Presentation of selected German and international (with special emphasis on local Peruvian) circumstances and available case studies on energy, water and material efficiency (technologies) according to specific client's needs. The **focal points** will be laid on demand-side management, heat and cooling energy, refrigeration system, Clean-in-

place (CIP) systems, and water and wastewater. Additionally, the energy and cost efficiency opportunities in cross-sectional technologies such as lighting, compressed air, hydraulic pumps and motors will be presented.

3. Presentation of research (field studies, trials, etc.) for significantly improving energy and material efficiency as well as utilisation of renewable sources (in-house resources) for power generation (e.g. from photovoltaic, wastewater or waste biomass) and further process energies (compressed air, heat, cold) as valuable measurement to switch fuel supply from fossil to renewable.

The aim of the **ICM** is to introduce the IfaS team and communicate the methodology and scope of the project to the business representatives, as well as to define the necessary data and organise the data collection process in the adjoining site visits.

Within the ICM, the available data (as well as the data accuracy) will be discussed and the strategy for the closure of data gaps will be defined. The data verification (by sampling and on-site observation), as well as additional data collection, will be done in the consecutive physical site visits of the production process and the plant sites by the consultancy team. At the end of the ICM and on-site visit phase, the acquired data will be presented within an interim presentation and report (**"Wrap-Up Meeting"**) on MFA results and shall be approved by the client for any further action within the consultancy service. The Wrap-Up Meeting should take place at the Client's premises to facilitate an intense discussion on the developed project ideas and consultancy results. It should serve as well to plan the following steps.

Based on the Wrap-Up Meeting, the need for additional specific data will be outlined and communicated to the business by the Client's assigned contact person in order to reach a mutual agreement among all project stakeholders upon the project opportunities, the following steps and required data to close gaps. If data gaps cannot be closed within acceptable time periods or at acceptable costs for the business (e.g. by additional measurements and/or laboratory analysis) the consultancy team will draw on appropriate literature default values identified by desk research.

Based on desk research and institutional experiences, the consultant will elaborate economic pre-feasibility studies of the technical optimisation strategies. Once the business planning of the designated improvement projects is completed, a draft version of the **"Standardised Report"** with all the relevant information will be made available to the Client as preparation for the **"Final Presentation & Conclusive Conference Call"** in order to define an implementation plan and **recommended follow-up** studies.

The following table exemplifies the tentative project duration for proposed REC. Based on previous projects and gained experiences it seems realistic to perform the REC during a period of 2 months, depending on the level of pro-active participation of the client and the data availability.

Table 2: Action plan for each individual REC

Action plan for the proposed Resource and Energy Efficiency Check (REC)									
Month		1				2			
Week		1	2	3	4	5	6	7	8
Task 1	Preparation of the REC								
Task 2 & 3	Material and Energy Flow Analysis and technical site visit to the Coishco Plant								
Task 4	Net-zero transformation pathway based on identified optimization measures								
Task 5	Reporting								

The site visit of the IfaS consultants can be carried out from October/November 2023. The date will be agreed with the client.

2.4 Task 1: Preparation of REC advisory services and onsite visit

Austral Group S.A.A., through its Energy Efficiency Committee, had already initiated the identification, evaluation and prioritization of energy efficiency measures that will enable the company achieve savings of energy costs and reduction of GHG emissions.

Nonetheless, the client seeks for external advisory for the diagnosis of the identified opportunities, analysis of the possible impacts in energy consumption and GHG emissions, and the development of a strategic roadmap which can prioritize the necessary investments and project implementations within the set period by the Energy Efficiency Committee.

The preparation phase is strongly dependent on the provision of required data, studies and previous works made by the client, as it consists mainly on the desktop review of the available data. Therefore, the main information request includes:

- Based on the questionnaire design by IfaS, the company internal data collection and records on the production processes and cross-sectional technologies, including quantities, seasonal availabilities, current utilisation degree and flow diagrams.
- List of energy efficiency measures previously identified.
- Evaluation matrix and its prioritization indicators.
- Previous studies to determine the corporate carbon footprint.
- Energy consumption, identified significant uses of energy, and seasonal variations.

During the preparation phase, IfaS will provide support to the client for the completion of the questionnaire for the initial analysis of the energy consumption behaviour and the initial efficiency check of cross-sectional technologies. After receiving the questionnaire, the consultant will analyse the provided data regarding their consistency, accuracy and completeness, and the findings will be presented in the ICM at the beginning of the site visit. Additionally, based on the desktop review of available documentation, the support will provide recommendations to close information gaps and preparation ad-hoc works that the client deems necessary, for instance, energy audit or carbon inventory.

2.5 Task 2: Analysis of the status quo

IfaS will employ its Material Flow Analysis (MFA) methodology, developed for numerous international Circular Economy and Zero Emission projects, where existing material and energy flows (water, resources, raw materials, energy, wastewater, waste etc.) through the production process and entire value chain of the client (from cradle to exit of the company site) will be determined based on qualitative and quantitative evaluation criteria. The process, methodology and necessary audit of the MFA are explained to the client during the ICM. The results of the initial analysis and desktop review of the provided will be presented during the ICM to determine the further data need as well as the onsite data compilation efforts during the onsite visit.

The gathered results will serve as an "MFA starting balance" which will be discussed during the ICM with key staff of the client onsite in order to timely start the additional data collection and sampling, as well as to verify the submitted data against the first onsite impressions.

The IfaS consultant team will concentrate the onsite evaluation and technical visit efforts on the Coishco processing plant.

The status quo analysis of the MFA phase finally results in a comprehensive mapping of (material, water and energy) flows and levelised cost of respective services (LCoS) for the entire process including in[2]- and output[3] balances with the main goal to define consumption hotspots. Consequently, the business shall be able to focus its resource and energy efficiency measures primarily on processes with intense consumption/demand (either in absolute consumption or absolute operational cost terms) in order to determine the priority areas for the further evaluation and potential/project development phase. Given the scarcity and the quality ('constraints') of resources, the mapping will also include the analysis of medium- to long-term seasonal variations on both, supply and demand sides, as well as relevant local national and/or other quality standards and legal compliance issues. This task will follow a stepwise procedure for the **quantitative and qualitative assessment of material flows.**

The quantitative MFA includes:

- Visualise the flow ('mapping') of all energies and materials within the existing processes stemming from the audit which necessarily include the process flows and also the required quality parameters based on the local standards; break-up of existing processes and flows in order to identify consumption hot-spots, failures, mismatch and inaccuracies, etc.
- Determine the specific consumption of the key processes (and subsequently compare and contrast these with national/international industrial benchmark values);
- Evaluate the price and embedded costs for resource and energy use to determine the levelised costs of service on a full cost accounting basis;

[2] Input: energy (electricity, gas, oil, fuels, etc.), raw material (water, auxiliary, etc.)
[3] Output: wastewater, emissions, products, waste streams

- Quantify the saving potential ('potentially for efficiency improvement') and also identify the required respective technological amendments/implementation;
- Quantify the medium- and long-term seasonal fluctuations of supply/resource availability and compare and contrast that with the forecasted demand (at least for the "Business as Usual Scenario" and the "Efficiency Improved Scenario") to identify if there would be a need for additional supply, which can be substituted by alternatives.
- Product-Residue-ratio analysis to determine the available waste and/or wastewater potential at the production sites. The water consumption and waste generation are analysed in the production plant and other activities of the plant (e.g. harbour, restaurants etc.).

Despite the current production volumes, the analysis takes future expansion plans and new product (lines and manufacturing practices) into account.

The qualitative MFA includes:

- Description of current management practices (e.g. organisational aspects, recording, monitoring and assessment of relevant data etc.);
- Assessment of the present conditions and state of the resource and energy-related instrumentation/infrastructure, inclusive of their maintenance practices;
- Assessment of the last efficiency project(s) undertaken and/or any predicted/proposed efficiency improvement measures for the next 24 months.
- Identification of critical points for metering and monitoring systems.

The key challenge during the onsite meeting is the verification and validation of data (sources) and the closing of data gaps. Wherever records and/or measured data are not available, the consultant will use assumptions based on experience and/or literature default values and communicate the assumptions directly with the client in order to ensure a smooth process onsite.

The key deliverables of this task are:

1. Presentation of the key findings of the MFA as the 'Resource (Material and Water) and Energy Map' as the main deliverable, which details on current consumption levels, masses, flow directions, interconnections, dependencies and associated LCOS, saving potentials, necessary investments for equipment (technology);
2. Estimation of (process) related CO_{2e} emissions (Baseline) for scope 1 (direct emission onsite, e.g. stationary combustion) and scope 2 emissions (indirect emission, e.g. grid electricity) and GHG starting balance.
3. Recommendations for metering and monitoring concept.

2.6 Task 3: Identification of additional opportunities to improve resource and energy efficiency

Based on the MFA results, especially in reference to the resource and energy map(s), the consultant will elaborate technical pre-designs and evaluate these efficiency measures based on relevant saving technologies or behavioural change. Finally, the economical pre-feasibility analysis and the evaluation of environmental impacts are elaborated by the consultant team. The objective of this task is to analyse the identified potentials of the business (as well as potentials from businesses in the vicinity which could be acquired by cross-company and/or company-regional authority cooperation) in order to define material and energy (thermal and electrical) re-use strategies within the companies' processing system.

The task will place a special focus on the reduction of fossil fuels as energy source for the steam boiler system and thermal energy provision.

Based on the available quantities and qualities of the potentials, the technical and economical pre-feasibility of the following **areas** will be analysed:

- Equipment

Identification of most important energy consumers (special regard to the refrigeration system and steam generation and supply), in order to define consumption "hot spots" and the associated technology used, including the operational management practice of the plant as well as the state of maintenance and depreciation value of the equipment. The equipment as a whole as well as of its spare parts are analysed, e.g. at pumping systems, both the pump and the drive engine performance are analysed.

- Production processes

Analysis of production processes, in order to identify efficiency measures through technical innovation or change of operational behaviour as well as synergies between production steps, e.g. utilisation of excess heat. The production process analysis will be used to double-check the provided data and data quality as well to evaluate the overall energy management system in terms of data recording and automation.

- Process management and operations

Adjustment of applied technologies throughout fine-tuning of plants, e.g. hydraulic balancing, pressure and flow adjustment, buffering and alignment of temperature level.

- Heat integration and recovery

Analysis of waste heat recovery options based on the identification of exhaust heat sources (e.g. compressed air productions, boilers and kilns, etc) and matching heat sinks in dependency on the required temperature level. Further investigation for exhaust heat utilisation via adsorption chilling for cooling processes. Optional exhaust heat/cold selling to neighbouring industries or residential areas via district cooling or heating systems.

- Raw and auxiliary material consumption

Check-up of raw material consumption, starting with the identification and evaluation of substitutes, via logistics within the business and utilisation in the process (comparison with similar companies) through to the recycling of waste from raw products.

- Energy and utility supply

Investigation of the business internal energy (including all forms of energy such as thermal -heat & cold-, steam, electricity) generation, distribution and usage, as well as water extraction and usage. Existing back-up systems will be analysed, too, in order to see if a constant operation could increase the overall system performance.

In the energy analysis the different energy tariff systems (capacity, work and times) will be analysed to determine of systemic demand side (and load) management could lead to internal energy cost saving by different energy tariff structures.

- Implemented Management Systems

Analysis of the currently employed (energy and environmental) management systems on-site. This entails a short description of the actions and measures taken in place, the organisational setup and the current capabilities. Additionally, the critical control points and significant energy users will be reviewed for the definition of specific metering and measurement points for data acquisition.

- Infrastructure

Evaluation of insulation standards for plants, units and buildings in order to develop strategies to decrease energy losses and further environmental burdens such as noise, odour and temperature to improve working conditions.

- Integration of renewable energy sources

Identification of untapped renewable energy potentials such as solar, wind, geothermal and biomass (organic waste) in order to switch fuel from fossil-based generation towards renewables. Final integration check of identified renewable energy options into the existing process structure, such as full, part or back-up supply.

Based on the MFA results, the consultancy team pre-designs and evaluates resource and energy efficiency options for the aforementioned areas. The objective is both to check the integration (adaptability) of the proposed action (technology) into the running processes of the business and to determine the Levelised Costs on a full cost accounting basis. Hence, it will be possible to compare the production cost of a proposed efficiency measure with the current business activities. Within this context, the following sub-steps for technical pre-feasibility of appropriate system improvements will be considered:

- Determination of energy or resource saving potential in kW, kWh, m3 or ton for identified energy consuming and material inefficiency hot spots;
- Prediction of designated thermal (heat, cold, steam) and/or electrical work for designated technology and definition of internal utilisation;
- Definition of capacity factors (installed capacity) for designated technologies;
- Definition and determination of logistical and (energy) storage requirements of designated concepts;
- Assessment of useful output and definition of internal re-utilisation;
- Acquiring a quotation for different scenarios from local technology provider (or international if locally not available), including the initial investment and operational costs.

The deliverables of this task are:

1. Wrap-up presentation (at the last day of the onsite visit) of the key findings for the material and energetic re-use of waste raw material, with details on electrical and thermal energy potential as well as necessary investments for equipment (technology);
2. A comprehensive report on the opportunities for efficiency improvement (including the reduction of consumption and the reuse of resources), the respective technologies, management practices, and standard amendments required and their corresponding cost-benefit analysis;
3. Report with input-output analysis, Levelized Costs of different end-energy forms (steam and/or heat/cooling price, electricity), materials (water, etc.) including a short technical description and a reference for a technology service provider;

Finally, the **economic pre-feasibility analysis** and the **evaluation of environmental impacts** are conducted by the consultant for each identified hot spot of the efficiency measurements.

2.7 Task 4: Development of net-zero transformation pathway

In order to visualize the net-zero pathway towards a Paris-aligned 1.5° goal, the different energy efficiency measures, and fossil-fuel switch toward green electricity powered modalities will be ranked and structured in short- (until 2025), mid- (until 2030), and long-term targets (until 2040) based on their likely economic development and preferability.

In the case that improvement projects and/or technology options, which do carry substantial impact on the net-zero transformation are not economically feasible at the time of reporting, it is essential to continuously evaluate existing technology improvement project and capitalize them once they provide an economic window of opportunity. Nevertheless, the nature of the consultancy services of technical support includes a foresight analysis and long-term perspective.

The relative GHG reduction potential is contrasted to the Business-as-usual (BAU) scenario and graphically visualized. Additionally, the weighted average carbon intensity of the production is calculated and illustrated.

The task includes the management of unavoidable carbon emissions to balance and offset the client's residual GHG emissions, which remain after a consistent implementation of energy and resource efficiency measurements. Different alternatives are existent and need to be evaluated carefully, in particular in the light of the clients ESG. In order to minimize associated climate risk and increase resilience in particular agri-business shall act as carbon sink, through the improvement of stored carbon within agricultural soils (in-setting). If no other options than acquisition of carbon removal certificates or development are existent a timely management of these acquisition in line with NDCs, SDGs and ESGs shall be envisaged.

The main deliverables of this task are:

1. Report and visualization of the carbon balance and net-zero transformation pathway (decarbonisation plan) based on the feasibility of implementation, cost-benefit effectiveness, and degree of impact for the company of the identified optimization measures.

2.8 Task 5: Preparation of a standardised report

Based on the results of the status quo analysis, the identification of opportunities for resource and energy efficiency optimisation and the wrap-up meeting on-site, the consultant will elaborate a standardised report of the findings.

The report shall include the methodology applied, the main assumptions, calculations performed, the actions recommendations and rough estimations regarding cost-benefit opportunities, energy savings and GHG mitigation potential.

A draft of the report will be sent to the client for its discussion to close information gaps and adjust pertinent changes for the final version of the report.

The REC shall be concluded with the final conference between the consultant team and the client. The aim of this follow-up call is to discuss the results presented in the standardised report and the possibilities for implementing the recommended actions and their integration into the Circular Economy Strategy of the client.

The deliverables of this task are:

1. Summary of the opportunities identified with the corresponding cost-benefit analysis;
2. Business plans and discounted cash flow analysis (including determination of initial investment, CAPEX, OPEX and Key Performance Indicators [KPI's]) for each hotspot;
3. Evaluation of CO_{2e} abatement potential for the developed scenarios;
4. Ranking of the most appropriate efficiency options in accordance to the economic impact with information on the technology as well as recommendations for follow-up assessments (in-depth analysis and/or implementation support) to elaborate the full technical and economic feasibility.
5. Recommendations for implementation roadmap (2030 and 2050) of improvement potentials on renewable energy, energy efficiency and GHG emission reduction.

3 Working Language

The working language of this undertaking shall be English. All official communications of this project (E-mails, communiqués, reports, presentations and the like) shall be entirely done in English. Internal communication, data provision and document sharing may be in Spanish.

4 Services of the Client

The client agrees to provide any and all data, information and the like pertinent to this project proposition/undertaking, free of charge, to the contractor (i.e. IfaS). Should the need be, the contractor may sign a non-disclosure agreement (NDA) upon the request of the client pertaining to the data/information procured.

The client will appoint an employee as the designated contact person who would liaise with the contractor on a regular basis to effectively execute this undertaking. Should the need be, this *liaison* will also support the contractor in compiling necessary documents and in the dissemination of information as well as in the coordination of meetings and discussions. This includes, in particular, the activities of coordination—such as in the case of surveys, data procurement, facility visits, interviews and the like—between the stakeholders of the client where the contractor has no control over.

5 Cost Structure of IfaS' Professional Services

The budgetary breakdown/cost structure of the professional services provided by IfaS/the contractor is tabulated in Table 3 below.

Table 3: Cost Structure and Proposal's Budget

Cost position			Man Days	Sum
			IfaS International Consultant	
		Daily rate	1,100.00 €	
Personnel Costs: REC advisory services of IfaS				
Task 1. Preparation of the REC advisory services			3	3,300.00 €
Task 2 & 3. Energy and Material Analysis and technical site visit For two IfaS consultants incl. 2 travelling days			8	8,800.00 €
Task 4. Net-zero transformation pathway based on identified optimization measures			2	2,200.00 €
Task 5. Reporting			7	7,700.00 €
Sub-total (net)			20	22,000.00 €
Tavel Costs (Lump Sum)	**€/Unit**	**Number**	**Consultants**	
International flights	2,800 €/Flight	1	2	5,600.00 €
Carbon offsetting of international flights	100 €/Flight	1	2	200.00 €
Accomodation	160 €/day	5	2	1,600.00 €
Daily allowance	60 €/day	5	2	600.00 €
Zwischensumme (netto)			8	8,000.00 €
Gesamtsumme (netto)			**28**	**30,000.00 €**

NOTE: This service is not taxable in Germany according to §3a Abs. 4 nr.3 UStG. Therefore, the tax liability remains with the recipient of the service (a.k.a. *Reverse Charge*).

In addition to the work packages specified herein, should the need be, the contractor may undertake ***additional*** activities such as coordination meetings, discussions, dissemination events, etc. at client-specified locations which may require travel, preparation and the like. The expenses incurred as a result will be charged/billed additionally to the client.

10 % of the net project cost may be used as payment of research allowances to professors entrusted with the work in the remuneration class W. The research allowances will be paid as a one-time payment during the duration of the project. The amount of the research allowances for project participants is decided by the president of Trier University of Applied Sciences, depending on the proportionate contribution to the project in accordance with the legal and internal university regulations.

6 Terms of Payment

The total professional service fee of IfaS is estimated at **30.000 EUR (net)**. By accepting this proposal, it is agreed to follow the two-step payment procedure that includes:

I.) a 50 % up-front payment of the total professional service fee within 14 days of the commissioning of the project

and

II.) the balance 50 % of the professional service fee payable within 14 days after the conclusion of the consultancy services by accepting the final report.

These payment terms can be amended upon mutual agreement between the client and the contractor.

7 Applicable Law

Under the governing law of the country of the contractor, this offer and its terms are subjected to the applicable law and regulations of the Federal Republic of Germany.

8 Other Provisions

Any amendments to this offer shall be in writing and signed by the client and the contractor. If any provision in this offer should be wholly or partially ineffective, the remainder of the provisions remains in force and binding for the partners (i.e. the client and the contractor). In this case, as required and appropriate, the partners undertake to replace the ineffective provisions by effective ones which may come as close as possible to the purpose of the ineffective ones.

The client and the contractor commit to taking any and all measures to ensure that all individuals involved in the subsequent undertaking of the proposition herein carry-out their work in a responsible and efficient manner whilst strictly adhering to the any and all confidentiality terms pertinent hereto. The nature of information regarded as such neither party disseminates nor pass any and all such information to a third party nor use it without the prior written consent of the partner.

9 Commitment Period & Ratification

Unless otherwise amended and/or ratified before the expiration, the offer herein is valid for a maximum of 30 calendar days from the date specified below.

Birkenfeld, May 11, 2023

Professor Dr. Dorit Schumann
President
Trier University of Applied Sciences

10 Acceptance of Offer & Commissioning

Place, Date

Full legal name of the signatory

Designation of the signatory

Institution